

High-performance perovskite/silicon tandem solar cells enabled by multifunctional titanium oxynitride recombination layers

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Monolithic perovskite/silicon tandem solar cells offer a promising pathway to surpass the efficiency limits of single-junction photovoltaics. However, their performance, stability and scalability are constrained by the recombination layer, which needs to simultaneously enable efficient charge recombination, high optical transparency and robust interfacial chemistry. Existing indium-containing transparent conductive oxides raise concerns regarding cost and sustainability, whereas silicon-based tunnel junctions suffer from parasitic optical losses. Here we show that titanium oxynitride (TiO_xN_y) can serve as a multifunctional, indium-free recombination layer that reconciles these competing requirements. Conductive TiO_xN_y enables efficient vertical carrier recombination, suppresses lateral leakage and provides anchoring sites for self-assembled monolayers (SAMs) via a tridentate binding configuration. As a result, we achieve power conversion efficiencies (PCEs) of 33.3% for 1.0-cm² devices and 30.6% for industrial-size (207.87 cm²) tandems, with enhanced operational stability. Our results establish TiO_xN_y as a scalable and sustainable interconnection strategy for tandem photovoltaics.

Perovskite solar cells (PSCs) have rapidly emerged as transformative photovoltaic technology owing to their exceptional optoelectronic properties and potential for low-cost fabrication^{1–3}. Single-junction PSCs have already achieved a certified PCE of 28%, approaching the Shockley–Queisser limit^{4,5}. To surpass this ceiling, perovskite/silicon (perovskite/Si) tandem solar cells have become one of the most promising routes^{6–9}. By combining the complementary absorption of wide-bandgap perovskite and silicon, they can theoretically achieve efficiencies beyond 43% (ref. 10). Perovskite/Si tandems have already reached certified PCEs of 35.2% (1.0 cm²) and 34.3% (industrial-size 261.0 cm²), demonstrating their strong commercial potential⁵.

A critical component of perovskite/Si tandems is the recombination layer (RL), which strongly influences PCE and stability^{11–13}. An ideal RL should enable efficient carrier recombination with minimal optical losses, provide a suitable surface for self-assembled monolayer (SAM) anchoring and maintain chemical and mechanical robustness during fabrication and operation^{14–16}. Current high-performance tandems predominantly employ indium-containing transparent conductive oxides (TCOs) or silicon-based tunnel junctions as the RL (Supplementary Table 1). TCOs offer high transparency and conductivity, but their high lateral conductivity can aggravate leakage from shunting defects, lowering fill factor (FF) and open circuit voltage

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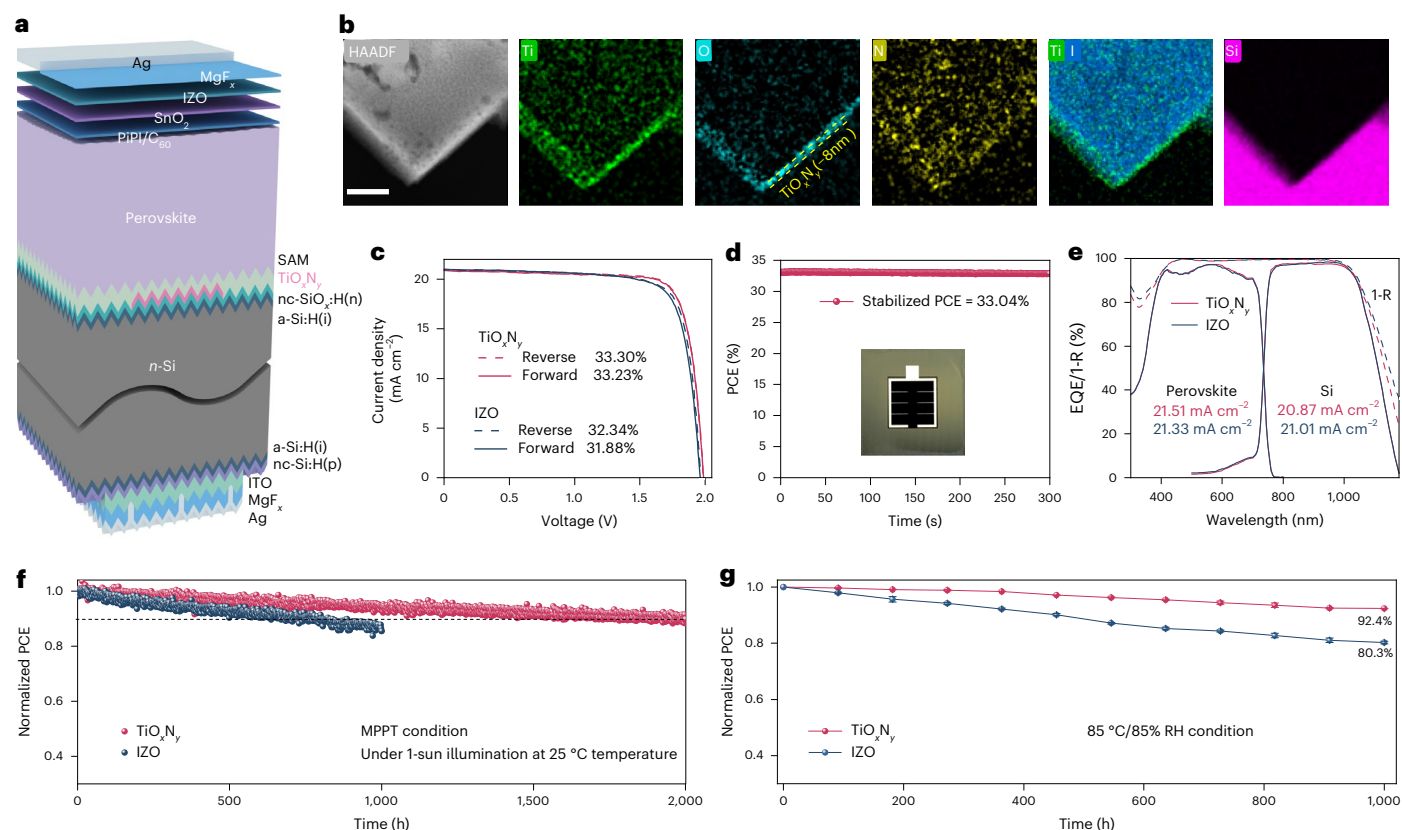


Fig. 1 | Performance of laboratory-size tandems with TiO_xN_y . **a**, Sketch of the tandem with TiO_xN_y , **b**, HAADF-STEM images with EDX mapping of TiO_xN_y and perovskite. Scale bar, 200 nm. **c**, Illuminated J - V curves of the champion devices with TiO_xN_y and IZO RLs. **d**, Stabilized power output of the champion device. Inset: photo of the 1- cm^2 tandem device. **e**, EQE and reflection curves of the tandem with TiO_xN_y and IZO RLs. **f**, MPP tracking under 1-sun and 25 °C with nitrogen atmosphere of the encapsulated tandems (testing protocol ISOS-L-II).

The initial PCE for the tandem employing TiO_xN_y /IZO RLs is 30.45% (28.71%). The dashed horizontal line indicates 790. **g**, PCE evolution of encapsulated tandems subjected to damp-heat stability test under 85 °C and 85% relative humidity (testing protocol ISOS-D-3). The symbols represent the mean values of $n = 3$ independently fabricated devices. Error bars represent the standard deviation. The curves are guides to the eye. PIPI, piperazinium iodide.

(V_{oc})^{12,15}. Silicon-based tunnel junctions suffer from parasitic absorption that limits the short-circuit current density (J_{sc})^{8,17}. These limitations motivate the development of indium-free RLs based on low-cost elements with tailored optoelectronic properties for tandem integration¹⁸.

Here we develop titanium oxynitride (TiO_xN_y) as an indium-free RL for monolithic perovskite/Si tandem solar cells. Deposited by scalable reactive sputtering, TiO_xN_y enables efficient carrier recombination, robust SAM anchoring and favourable perovskite crystallization. We further investigate its composition, optoelectronic properties, recombination mechanism and interfacial chemistry and demonstrate high-efficiency tandems at both laboratory and industrial scales.

Performance of tandem solar cells

Figure 1a shows the architecture of the perovskite/Si tandem solar cells with TiO_xN_y RLs. The silicon heterojunction (SHJ) bottom cell features double-side submicron textures, which reduce reflection and facilitate conformal layer deposition. Reactive sputtering produced a uniform TiO_xN_y RL on the textured surface. High-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) and elemental mapping by energy-dispersive X-ray spectroscopy (EDX) confirm the homogeneous distribution of Ti, O and N across the interface (Fig. 1b). A continuous TiO_xN_y layer (~ 8 nm) was identified the wide-bandgap perovskite absorber and subsequent functional layers, including C_{60} , SnO_2 , indium zinc oxide (IZO) and MgF_x (Supplementary Fig. 1). Titanium (Ti) and iodine (I) partially overlap in EDX spectra because of their similar emission energies and limited detector resolution. Cross-sectional scanning electron microscopy (SEM) further confirms

a compact, large-grain perovskite layer with well-defined interfaces on the textured layers (Supplementary Fig. 2).

The performance of laboratory-size tandem with optimized TiO_xN_y RL is shown in Fig. 1c, alongside a control device employing conventional IZO RL (10 nm). The TiO_xN_y -based tandem achieves a champion PCE of 33.3% (1 cm^2), with a V_{oc} of 1.988 V, J_{sc} of 20.91 mA cm^{-2} and FF of 80.11%. Stabilized power output confirms a PCE of 33.04% (Fig. 1d), whereas independent certification by the Japan Electrical Safety and Environment Technology Laboratories verified a PCE of 32.8% (Supplementary Fig. 3). In comparison, the IZO-based control exhibits a lower PCE of 32.34%, mainly due to reduced V_{oc} (1.965 V) and FF (78.41%). TiO_xN_y RL also delivers a higher average PCE (32.77% versus 31.22%) and improved reproducibility (Supplementary Fig. 4). Suns- V_{oc} analysis reveals lower selectivity and transport loss in TiO_xN_y -based devices, indicating improved interfacial charge extraction and reduced series resistance, which contribute to gains of 23 mV V_{oc} and 1.7% absolute in FF (Supplementary Fig. 5). In addition, voltage-dependent shunt resistance (R_{sh}) measurements show that TiO_xN_y maintains 30% higher R_{sh} with weaker voltage dependence than IZO (Supplementary Fig. 6), consistent with suppressed leakage pathways and improved charge collection. External quantum efficiency (EQE) spectra (Fig. 1e) yield integrated photocurrent densities of 21.51 mA cm^{-2} for the perovskite subcell and 20.87 mA cm^{-2} for the SHJ subcell, matching the measured J_{sc} . Although TiO_xN_y introduces slightly lower near-infrared EQE due to parasitic absorption (Supplementary Fig. 7), its higher R_{sh} mitigates lateral shunting and current leakage, resulting in a higher FF.

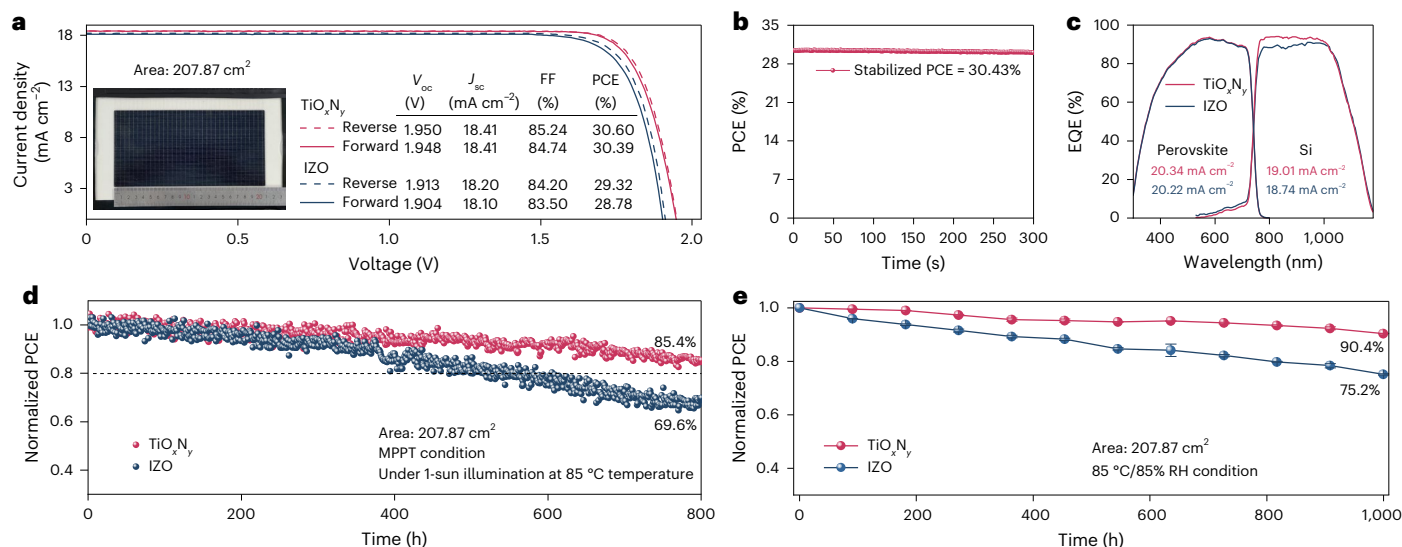


Fig. 2 | Performance of industrial-size tandems with TiO_xN_y . **a**, Illuminated J - V curves of champion device with TiO_xN_y and IZO RLs. Inset: photo of the tandem mini-module (207.87 cm²). **b**, Stabilized power output of the champion tandem. **c**, EQE of the champion tandems with TiO_xN_y and IZO RLs. **d**, MPP tracking under 1-sun and 85 °C with nitrogen atmosphere of the encapsulated tandems (testing protocol ISOS-L-2I). The initial PCE for the tested tandem

employing TiO_xN_y /IZO RLs is 29.47% (27.95%). The dashed horizontal line indicates 790. **e**, PCE evolution of encapsulated tandems subjected to damp-heat thermal stability test under 85 °C with RH of 85% (testing protocol ISOS-D-3). The symbols represent the mean values of $n = 3$ independently fabricated devices. Error bars represent the standard deviation. The curves are guides to the eye.

Operational stability was evaluated by maximum power point (MPP) tracking under continuous 1-sun illumination in nitrogen at room temperature (Supplementary Fig. 8). Encapsulated TiO_xN_y -based tandem retains 90% of its initial PCE (790) after 2,000 h, outperforming IZO-based tandems (790 $\approx$ 900 h) under identical conditions (Fig. 1f). Decomposed MPP data indicate that the primary degradation pathway is voltage loss. The IZO-based device exhibits a pronounced decline in V_{mpp} associated with a reduction in R_{sh} from 7,590 to 1,830 Ω cm², whereas the TiO_xN_y -based device maintains a higher R_{sh} together with stable V_{mpp} and J_{mpp} throughout ageing (Supplementary Figs. 9 and 10 and Supplementary Table 2). The largely unchanged J_{mpp} suggests that shunting remains spatially confined at 25 °C despite the reduction in R_{sh} . Cross-sectional SEM images after ageing show that the TiO_xN_y -based device preserves continuous morphology without observable delamination, whereas the IZO-based device exhibits less distinct interfacial boundaries (Supplementary Fig. 11a,b). Post-ageing PL mapping reveals expanded quenched regions in the control device that correlate with interfacial defects, whereas the TiO_xN_y -based device maintains uniform, high-intensity emissions across the active area (Supplementary Fig. 11c,d).

We further evaluated encapsulated tandems under damp-heat conditions according to International Electrotechnical Commission protocol 61215 (85 °C/85% relative humidity). The laboratory-scale tandem retains 92.4% of its initial PCE after 1,000 h substantially outperforming the control device (80.3%) (Fig. 1g). Both devices exhibit pronounced losses in V_{oc} and FF with largely unchanged J_{sc} , indicating that degradation primarily originates from interfacial processes rather than bulk material decomposition (Supplementary Fig. 12 and Supplementary Table 3). Cross-sectional SEM after damp-heat ageing shows that the TiO_xN_y -based device maintains a continuous morphology, whereas the control device develops more pronounced interfacial irregularities and localized voids (Supplementary Fig. 13a,b). Consistently, PL mapping reveals quenched regions near the RL in the IZO device, whereas the TiO_xN_y device retains uniform, high-intensity emission across the active area (Supplementary Fig. 13c,d). These results suggest that the robust tridentate SAM on TiO_xN_y suppresses defect formation and shunting pathways, preserving device performance under both ageing protocols.

We further validated the TiO_xN_y RL in industrial-size tandem solar cells (210 mm $\times$ 105 mm), featuring a slot-die coated wide-bandgap perovskite top cell and a similar architecture. A certified PCE of 30.6% with negligible hysteresis was achieved for a tandem mini-module fabricated from a single industrial-size cell (207.87 cm²; Fig. 2a and Supplementary Fig. 14). Stabilized power output confirms a PCE of 30.43% (Fig. 2b). The slot-die coated perovskite exhibited optical, structural and microstructural properties comparable to those spin-coated films (Supplementary Fig. 15), whereas device reproducibility was evidenced by a narrower PCE distribution across multiple batches (Supplementary Fig. 16). Compared with laboratory-size tandems, the industrial devices shown a much lower J_{sc} of 18.41 mA cm⁻² owing to optical losses from encapsulation, screen-printed metal fingers and the absence of a rear reflector in the SHJ bottom cell (Fig. 2c and Supplementary Figs. 17 and 18). Nevertheless, TiO_xN_y exhibited better J_{sc} retention upon upscaling than IZO (2.50 mA cm⁻² versus 2.79 mA cm⁻² loss), consistent with improved device uniformity and interfacial robustness during large-area fabrication. The FF increased to 85.24%, partially compensating for the reduced J_{sc} . Simulation program for integrated circuits emphasis (SPICE) simulations indicate that the higher FF originates from reduced shunt losses and edge recombination, together with a current mismatch that shifts the operating point towards a higher-FF regime (Supplementary Fig. 19)¹⁹. The V_{oc} remained high at 1.95 V, comparable to that of laboratory-scale devices, highlighting efficient carrier recombination and high-quality slot-die-coated perovskite growth on TiO_xN_y . We further investigated the long-term stability of the tandem mini-module. Under continuous 1-sun illumination at 85 °C in nitrogen, the TiO_xN_y -based device retained 85.4% of its initial PCE after 800 h of MPP tracking, outperforming the control device (69.6%) (Fig. 2d). Unlike the voltage-dominated degradation observed at 25 °C, ageing at 85 °C resulted in losses in both J_{mpp} and V_{mpp} . The control device exhibited pronounced J_{mpp} oscillations and faster performance decay, whereas the TiO_xN_y -based device displayed only gradual decreases in both parameters (Supplementary Fig. 20). This behaviour is attributed to thermally enhanced dynamic shunting and ion migration, which reduce R_{sh} and the effective carrier collection area (Supplementary Fig. 21 and Supplementary Table 4). The high lateral resistance of TiO_xN_y suppresses filament propagation, resulting

in slower degradation. In a damp-heat test, the tandem mini-module retains 90% of its initial PCE after 1,000 h, markedly outperforming the control device (75.2%) (Fig. 2e). Similar to the laboratory-size device, degradation was primarily associated with reductions in V_{oc} and FF (Supplementary Fig. 22 and Supplementary Table 5). The improved stability is attributed to the robust TiO_xN_y /SAM interface, which suppresses non-radiative recombination, ion migration and charge leakage.

Optoelectronic properties of TiO_xN_y

Atomic-layer-deposited TiO_xN_y , featuring a low resistivity and a high electron concentration, has been demonstrated to be an efficient electron-transport layer for both silicon and organic solar cells²⁰. Sputtered TiO_xN_y was also demonstrated to be an efficient electron-transport layer in single-junction perovskite solar cells, achieving high fill factors (> 86%) and certified PCEs up to 22.6% on 1-cm² devices²¹. We first investigated and optimized the composition and optoelectronic properties of the sputtered TiO_xN_y , which was deposited under fixed N_2/Ar ratio, power and chamber pressure of 1%, 200 W and 1×10^{-4} Pa, respectively. High-resolution X-ray photoelectron spectroscopy (XPS) of Ti 2p and N 1s (Fig. 3a,b) reveals a mixture of Ti–O, Ti–N and Ti–ON bonding states, indicating the co-existence of TiO_2 , TiN and TiON phases. Deconvoluted N 1s spectrum further confirms the co-existence of nitride and oxynitride components. Quantitative analysis indicates elemental concentrations of 34.1% Ti, 39.9% O and 26.0% N, close to phase-stable stoichiometry of $\text{Ti}_{2.75}\text{O}_3\text{N}_2$ ²². Following convention, we denote as-deposited film with a non-stoichiometric chemical formula TiO_xN_y ^{23,24}. The optical bandgap of TiO_xN_y was determined to be 3.36 eV (Supplementary Fig. 23), which was close to the ALD TiO_xN_y ²⁰. Figure 3c shows the sheet resistance (R_s , which primarily governs lateral current transport) of TiO_xN_y as a function of thickness. The R_s decreases rapidly with increasing thickness, reaching a low R_s of 1,852 $\Omega \text{ sq}^{-1}$ (ohms per square) at 12 nm. This trend is accompanied by a reduction in average transmittance across 400–1,200 nm (Fig. 3d). Compared to the IZO (10 nm), the optimal TiO_xN_y RL (8 nm) shows a higher R_s (3,041 $\Omega \text{ sq}^{-1}$) and a lower average transmittance (particularly at the near-infrared regions), which can be ascribed to the high electron concentration of $1.57 \times 10^{22} \text{ cm}^{-3}$ and low mobility of $0.3 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ for TiO_xN_y (Supplementary Table 6). These properties arise from the quasi-metallic TiN phase embedded in TiO_xN_y , which contributes to high conductivity and carrier density²⁵. The reduced NIR transmittance probably explains the slightly lower J_{sc} observed in the TiO_xN_y -based tandems (Fig. 1c).

We further measured the contact resistivity (ρ_c) between the TiO_xN_y RL and SHJ bottom cell (Supplementary Fig. 24). TiO_xN_y exhibits a lower ρ_c (28 $\text{m}\Omega \text{ cm}^{-2}$) than IZO (45 $\text{m}\Omega \text{ cm}^{-2}$), facilitating vertical carrier transport and contributing to the high FF. Tandem performance is sensitive to TiO_xN_y thickness (Fig. 3e and Supplementary Fig. 25). Increasing thickness from 4 nm to 12 nm progressively improves FF owing to enhanced conductivity and reduced charge accumulation at the recombination interface. However, TiO_xN_y exhibits a non-negligible extinction coefficient across the measured spectrum (Supplementary Fig. 26), resulting in reduced transmittance with increasing thickness (Supplementary Fig. 27). Compared with highly transparent IZO, thicker TiO_xN_y layers introduce greater parasitic absorption, necessitating a trade-off between electrical and optical performance. Consistent with this behaviour, reflectance decreases with increasing thickness as more incident light is absorbed within the film (Supplementary Fig. 28). EQE measurements further reveal a reduction in SHJ bottom-cell photocurrent, whereas the perovskite top cell remains largely unaffected (Supplementary Fig. 29). The lower near-infrared transmittance of thicker TiO_xN_y leads to an EQE reduction near 1,000 nm (~15% at 12 nm), accounting for the slightly reduced J_{sc} observed in TiO_xN_y -based tandems. These results identify ~8 nm as the optimal TiO_xN_y thickness. To clarify the carrier recombination

mechanism, we determine the band diagram of the recombination junctions by measuring the work function (WF) using ultraviolet photoelectron spectroscopy (UPS). As-deposited TiO_xN_y exhibits a WF of 4.41 eV, comparable to literature values^{21,25}. After anchoring a mixed SAM (MeO-4PACz and 4PADCB), the WF increase substantially to 5.06 eV (Supplementary Fig. 30). Notably, the TiO_xN_y shows a larger WF shift (+ 0.65 eV) than that of IZO (+ 0.31 eV; Fig. 3f). According to the Helmholtz equation, this WF shift reflects a higher dipole surface density of SAM on TiO_xN_y ¹⁵. The higher WF facilitates the transfer of charge carriers at the perovskite/SAM/ TiO_xN_y interface, resulting in effective carrier recombination and higher tandem performance.

To compare interfacial band alignment within the full device, band diagrams of tandems employing TiO_xN_y and IZO RLs were simulated using SCAPS-1D based on experimentally measured layer properties (Fig. 3g–h and Supplementary Tables 7–9). The simulations were used to visualize band offsets and band bending rather than to provide quantitatively accurate energetics. At the recombination junction, efficient carrier extraction from both subcells is required for charge recombination^{26,27}. Without considering the SAM-induced interfacial dipole^{28,29}, the perovskite band bends downward, which is unfavourable for carrier extraction. This effect can be reproduced by introducing an effective interfacial defect density at the perovskite/SAM interface (Supplementary Fig. 31a). The simulations further indicate that replacing IZO with TiO_xN_y eliminates the energy barrier at the RL/nc-SiO_x(n) interface, facilitating electron extraction from the SHJ bottom cell. The high carrier density of TiO_xN_y may further enhance charge screening and lower the transport barrier, as supported by the increased barrier height when the electron density is reduced to 10^{20} cm^{-3} (Supplementary Fig. 31b). These results suggest that TiO_xN_y enables more efficient carrier extraction from both subcells, contributing to the improved tandem performance.

Adsorption of self-assembled monolayer on TiO_xN_y

In perovskite/Si tandems, the RL is crucial for enabling efficient electron-hole recombination via SAM-modified interfaces¹⁴. Conventional sputtered TCOs (for example, ITO and IZO) often exhibit low SAM coverage density and poor surface uniformity due to the scarcity hydroxyl (–OH) groups, which limits interfacial charge transfer, device performance and long-term stability. We first investigate the TiO_xN_y -based tandem performance using different SAMs. The best performance is obtained with a mixed SAM of MeO-4PACz and 4PADCB (Supplementary Fig. 32). This synergy arises because MeO-4PACz improves wettability while 4PADCB provides steric hindrance, reducing aggregation and favouring parallel molecular orientation relative to the RL surface³⁰. The resulting SAM layer probably forms a more uniform molecular assembly and enhances interfacial contact.

To further investigate the adsorption behaviour of MeO-4PACz and 4PADCB on both TiO_xN_y and IZO surfaces, we performed density functional theory (DFT) calculations. First, surface energy calculations established that the TiO_xN_y (100) facet exhibits the lowest surface energy (Supplementary Table 10), indicating it is the most thermodynamically stable facet for our subsequent slab modelling. Specifically, we analysed bidentate binding configurations involving Ti–O and In–O coordination interactions (Fig. 4a,b and Supplementary Fig. 33), which are known to play a crucial role in charge transport modulation and interfacial stabilization. Binding energy calculations further confirm stronger interactions between SAM anchoring groups and Ti sites than with In sites, reflecting the higher chemical reactivity of Ti in bidentate configurations (Supplementary Tables 11 and 12). Notably, a tridentate binding mode emerges when the SAM molecules interact with adjacent nitrogen elements within TiO_xN_y , yielding enhanced binding energies (–6.38 eV for MeO-4PACz and –6.53 eV for 4PADCB; Fig. 4c, Supplementary Tables 11 and 12), underscoring the critical role of nitrogen incorporation in facilitating robust chemisorption.

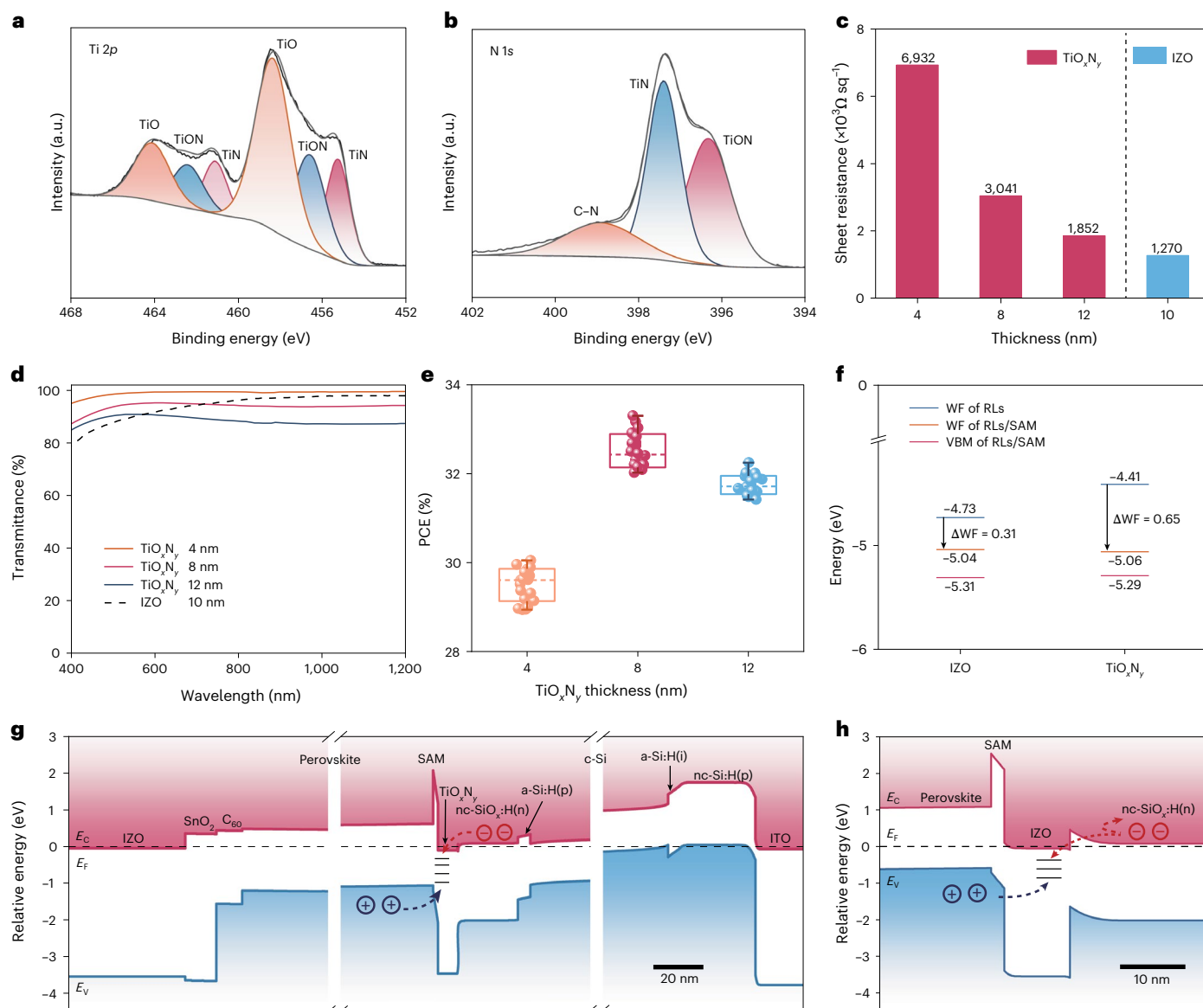


Fig. 3 | Optoelectronic properties of TiO_xN_y . **a, b**, XPS core-level spectra of Ti 2p (**a**) and N 1s (**b**) for sputtered TiO_xN_y . **c**, R_s of TiO_xN_y and IZO as a function of thickness. **d**, Transmittance spectra of TiO_xN_y with different thicknesses. **e**, Effect of TiO_xN_y thickness on the PCE of tandem solar cells. Boxplots show data from $n = 20$ independently fabricated devices per condition. Box boundaries represent the mean ± 1 standard deviation; the dashed line indicates the median; whiskers extend to the minimum and maximum values. Individual data points are overlaid

as scatter plots. **f**, Band position and work function of TiO_xN_y and IZO before and after SAM anchoring. **g, h**, Simulated relative energy band diagrams of tandem device with TiO_xN_y RL (**g**) and the perovskite/IZO/SHJ structure (**h**). E_c , E_f and E_v denote the conduction band minimum, Fermi level and valence band maximum, respectively. The x-axis scale is discontinuous to highlight the thin interfacial regions, given that the perovskite top cell and c-Si bottom cell are substantially thicker than the interfacial layers; omitted segments are not shown to scale.

Furthermore, these enhanced binding trends remain highly consistent when dipole corrections are accounted for (Supplementary Table 13). Crystal orbital Hamilton population (COHP) and integrated COHP (ICOHP) analyses further reveal that N–O interactions are considerably stronger than Ti–O bonds for both SAMs (ICOHP for N–O versus Ti–O: -4.24 eV versus -1.61 eV for MeO-4PACz; Fig. 4d; and -4.21 eV versus -1.60 eV for 4PACDB; Supplementary Fig. 34). These results highlight the critical role of nitrogen in enhancing chemisorption strength and providing additional active sites for SAM anchoring.

Surface potential uniformity of TiO_xN_y and IZO surfaces before and after SAM modification was then probed using Kelvin probe force microscopy (KPFM) (Fig. 4e,f and Supplementary Fig. 35). Bare TiO_xN_y exhibits much larger local potential variations than IZO. However, after mixed SAM anchoring, TiO_xN_y /SAM shows a lower average surface potential than IZO/SAM. These results align with the work function

changes, further confirming that the TiO_xN_y surface possesses a higher density of anchoring sites, which facilitates the preferential orientation and stacking of SAMs¹⁵. Conductive atomic force microscopy (c-AFM) shows that TiO_xN_y /SAM delivers a higher hole-injection current (13.2 nA versus 8.7 nA) and a lower electron-leakage current (7.3 nA versus 8.3 nA) than IZO/SAM (Supplementary Fig. 36), confirming a lower hole-injection resistance and a denser and more uniform SAM coverage³¹. This dual enhancement functionally evidences superior SAM quality on TiO_xN_y . DFT calculations (Supplementary Table 14 and 15) further support this trend, showing substantially larger work-function changes for SAMs on TiO_xN_y (0.530 eV for 4PACDB, 0.365 eV for MeO-4PACz) than on IZO (0.013 eV for 4PACDB, 0.012 eV for MeO-4PACz), in agreement with experiment (Supplementary Fig. 37). These results suggest that TiO_xN_y offers a higher density of chemically active anchoring sites, promoting better SAM coverage and improved surface potential

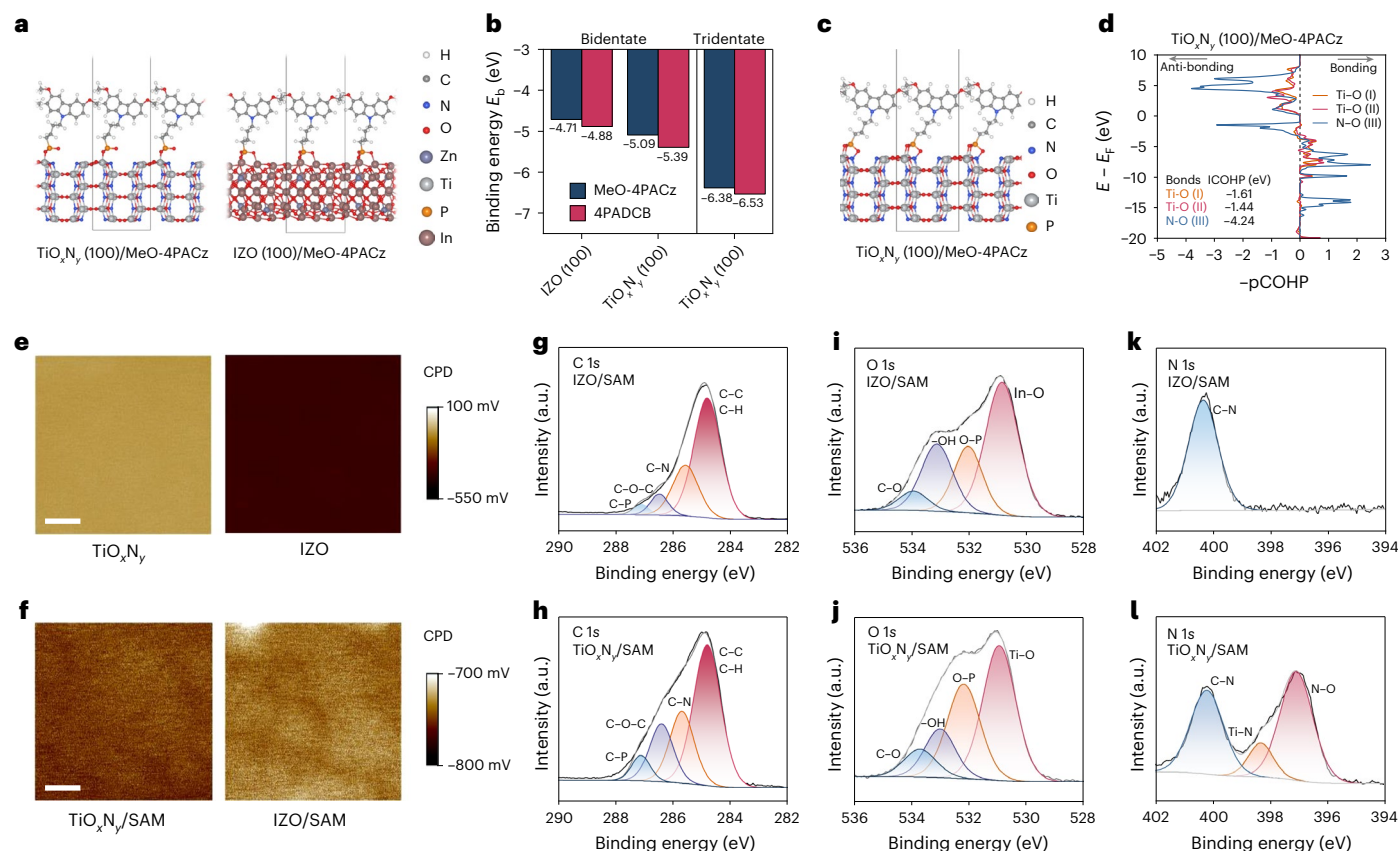


Fig. 4 | SAM adsorption on TiO_xN_y . **a**, Visual representation of bidentate binding modes for $\text{TiO}_x\text{N}_y(100)/\text{MeO-4PACz}$ and $\text{IZO}(100)/\text{MeO-4PACz}$ systems. **b**, Binding energies of MeO-4PACz and 4PACDB on the TiO_xN_y and IZO. **c**, Visual representation of tridentate binding modes for $\text{TiO}_x\text{N}_y(100)/\text{MeO-4PACz}$. **d**, COHP and ICOHP analysis of the $\text{TiO}_x\text{N}_y(100)/\text{MeO-4PACz}$ interface with bonding interactions labelled numerically. **e–f**, KPFM images of TiO_xN_y and IZO without (**e**) and with (**f**) SAM anchoring. Scale bars, 400 nm. **g–l**, XPS core-level spectra of C 1s, O 1s and N 1s for IZO (**g, i, k**) and TiO_xN_y (**h, j, l**) after SAM anchoring. CPD, contact potential difference.

interactions at the interface; positive values indicate bonding and negative values indicate antibonding. ICOHP values reflect bond strength. **e, f**, KPFM images of TiO_xN_y and IZO without (**e**) and with (**f**) SAM anchoring. Scale bars, 400 nm. **g–l**, XPS core-level spectra of C 1s, O 1s and N 1s for IZO (**g, i, k**) and TiO_xN_y (**h, j, l**) after SAM anchoring. CPD, contact potential difference.

homogeneity across the interface³². X-ray photoelectron spectroscopy (XPS) further confirms higher SAM coverage on TiO_xN_y compared with IZO. The C 1s spectra display three distinct components: a dominant C–C/C–H peak, a medium-intensity C–N signal and a weaker C–P contribution (Fig. 4g,h). The observed peak intensity ratios closely reflect the molecular composition of the mixed SAM, confirming successful deposition and structural preservation on both TiO_xN_y and IZO. Deconvolution of the O 1s spectra reveal four components (Fig. 4i,j): lattice metal-oxygen (In–O or Ti–O), phosphorus-oxygen (P–O), hydroxyl groups (–OH) and carbon-oxygen (C–O). Notably, a higher P–O/Ti–O ratio (0.71) is obtained for $\text{TiO}_x\text{N}_y/\text{SAM}$, compared to that of IZO/SAM (0.43), indicating a higher SAM coverage density and stronger interfacial bonding with phosphonic acid anchoring groups on TiO_xN_y ³³. High-resolution P 2p XPS spectra reveal a systematic decrease in binding energy from IZO to TiO_xN_y , confirming the most complete deprotonation and strongest tridentate covalent anchoring of the SAM on TiO_xN_y (Supplementary Fig. 38). Fourier transform infrared (FTIR) spectra further demonstrate that the P=O ($\sim 1,240\text{ cm}^{-1}$) and P–OH ($\sim 1,020\text{ cm}^{-1}$ and $\sim 970\text{ cm}^{-1}$) stretching peaks are prominent on IZO but nearly disappear on TiO_xN_y , indicating complete condensation and stronger covalent P–O–Ti bonding (Supplementary Fig. 39). Elemental ratios also confirm higher phosphorus content relative to Ti, and a reduced Ti 2p signal, suggesting denser SAM packing on TiO_xN_y that driven by the superior chemical affinity of TiO_xN_y for anchoring groups compared to IZO³⁴ (Supplementary Table 16). Distinct bonding behaviours are also observed in the N 1s spectra (Fig. 4k,l). For IZO/SAM, the signal primarily originates from C–N species in the SAM.

In contrast, $\text{TiO}_x\text{N}_y/\text{SAM}$ exhibits a reduced Ti–N signal and increased N–O component, implying preferential interaction between nitrogen atoms in TiO_xN_y and oxygen in mixed SAM, consistent with DFT calculations. Such interactions favour higher-order coordination (bidentate and tridentate bonding), further stabilizing the interface and enhancing SAM adsorption^{35,36}. Together, these results demonstrate that TiO_xN_y offers a chemically favourable platform for robust SAM anchoring, enabling denser coverage, stronger adhesion and more homogeneous surface potential. These interfacial improvements are essential for achieving high-efficiency, stable tandem solar cells.

Perovskite on TiO_xN_y self-assembled monolayer

The RL/SAM/perovskite interface plays a pivotal role in determining the nucleation and crystallization of wide-bandgap perovskite, which directly impacts optoelectronic properties and tandem performance. To probe the interfacial properties for perovskite nucleation, we first evaluated substrate wettability. Whereas using a mixed SAM of MeO-4PACz and 4PACDB, the contact angle of the perovskite precursor on $\text{TiO}_x\text{N}_y/\text{SAM}$ approached 0° , lower than the 9.3° observed on IZO (Supplementary Fig. 40). This near-perfect wetting indicates a thermodynamically favourable interface, due to a stronger interaction between SAM and TiO_xN_y surface, promoting denser and more ordered SAM packing³⁷. Buried-interface SEM images (Supplementary Fig. 41) show compact polygonal grains with sharp boundaries and excellent coverage on $\text{TiO}_x\text{N}_y/\text{SAM}$, in contrast to irregular grains with incomplete coverage on IZO/SAM. Although MeO-4PACz alone provides excellent wettability, its strong aggregation tendency limits device performance.

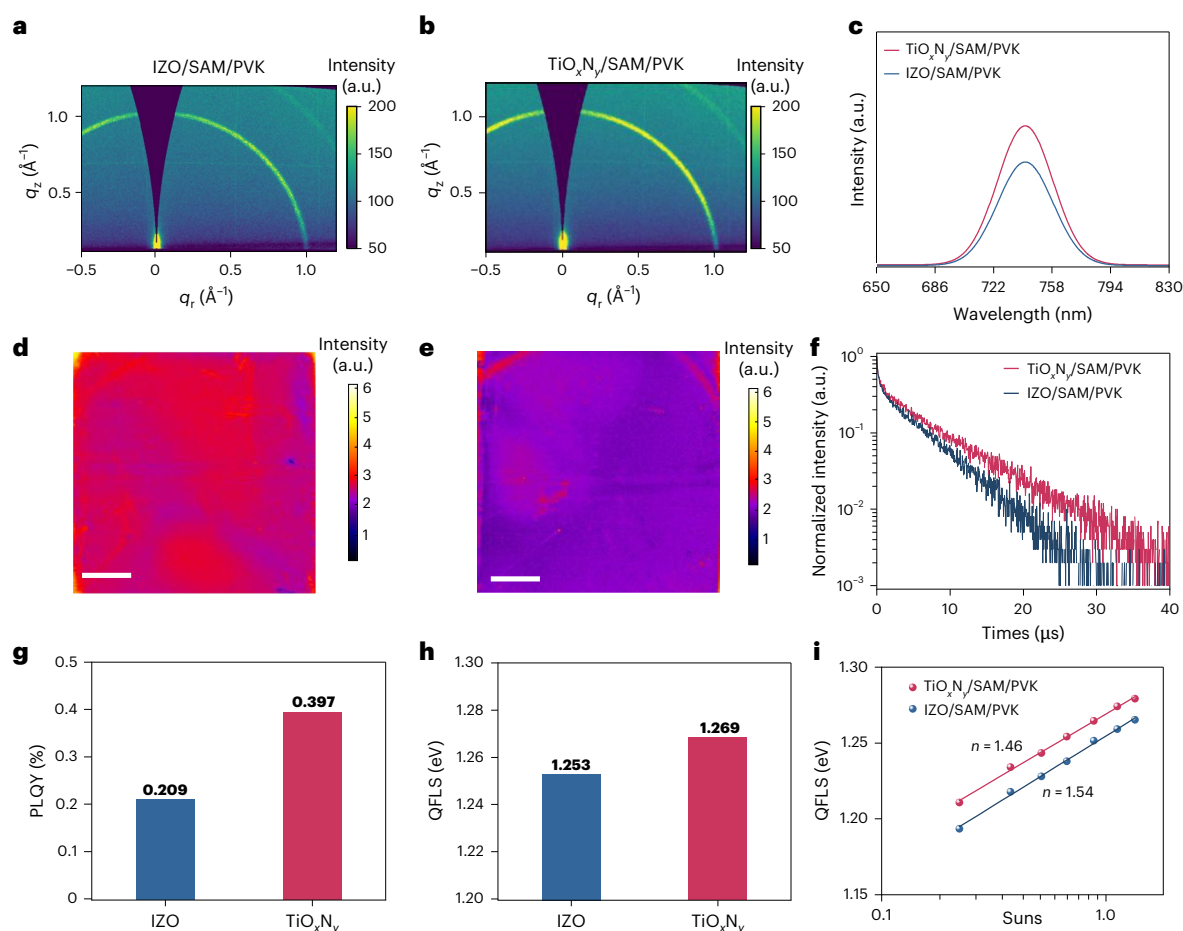


Fig. 5 | Perovskite grown on $\text{TiO}_x\text{N}_y/\text{SAM}$. **a,b**, GIWAXS images of perovskites grown on IZO/SAM (**a**) and $\text{TiO}_x\text{N}_y/\text{SAM}$ (**b**). **c**, PL spectra of perovskites grown on $\text{TiO}_x\text{N}_y/\text{SAM}$ and IZO/SAM. **d,e**, PL imaging for perovskites grown on $\text{TiO}_x\text{N}_y/\text{SAM}$ (**d**) and IZO/SAM (**e**). Scale bars in **d** and **e**, 0.5 cm. **f–h**, TRPL spectra

(**f**), PLQY (**g**) and QFLS (**h**) of perovskites grown on $\text{TiO}_x\text{N}_y/\text{SAM}$ and IZO/SAM. **i**, Intensity-dependent QFLS of perovskites grown on $\text{TiO}_x\text{N}_y/\text{SAM}$ and IZO/SAM. All the samples were prepared on textured silicon substrates. The solid lines indicate linear fits.

Grazing-incidence wide-angle X-ray scattering (GIWAXS) measurements reveal pronounced difference in perovskites grown on $\text{TiO}_x\text{N}_y/\text{SAM}$ and IZO/SAM (Fig. 5a,b). On $\text{TiO}_x\text{N}_y/\text{SAM}$, the perovskite exhibits a higher diffraction intensity at $q_r \approx 1.0 \text{ \AA}^{-1}$, corresponding to the (100) planes. This is further supported by X-ray diffraction (XRD), which shows a more intense (100) diffraction peak than that on IZO/SAM (Supplementary Fig. 42). This facilitates more efficient charge carrier dynamics and suppresses non-radiative recombination at the buried interface. Consequently, these structural improvements contribute to the substantially higher FF and V_{oc} achieved in the TiO_xN_y -based tandem solar cells³⁸.

Photoluminescence (PL) further highlights the superior interfacial quality of $\text{TiO}_x\text{N}_y/\text{SAM}$. Perovskites grown on $\text{TiO}_x\text{N}_y/\text{SAM}$ exhibit substantially higher PL intensity than those on IZO/SAM (Fig. 5c). PL mapping reveals homogeneous emission across $\text{TiO}_x\text{N}_y/\text{SAM}$, in contrast to heterogeneous and weaker emission observed for IZO/SAM (Fig. 5d,e), suggesting localized non-radiative recombination from defect clusters in the latter. PL images of the tandem solar cells with TiO_xN_y and IZO RL corroborate these findings (Supplementary Fig. 43). Time-resolved photoluminescence (TRPL) measurements further confirm the suppression of non-radiative recombination at the interface (Fig. 5f). To quantitatively resolve the competing interfacial processes, we performed differential lifetime analysis based on triple-exponential fitting, defined as lifetime $\tau = -\{d\ln[\Phi(t)]/dt\}^{-1}$, where $\Phi(t)$ is the time-dependent PL photon flux³⁹. This analysis reveals that the perovskite film on $\text{TiO}_x\text{N}_y/\text{SAM}$ not only shows faster

initial charge extraction kinetics but also achieves a markedly prolonged asymptotic lifetime exceeding 7.1 μs . In contrast, the IZO/SAM structure attains a maximum differential lifetime of only about 5.4 μs (Supplementary Fig. 44). These results indicate reduced interfacial trap density and carrier recombination loss enabled by uniform and chemically robust SAM anchoring on TiO_xN_y ⁴⁰. Absolute PL quantum yield (PLQY) of perovskites grown on different RL/SAM provides additional insight. The perovskite grown on the IZO/SAM yields a low PLQY of 0.209%, corresponding to a quasi-Fermi level splitting (QFLS) of 1.253 eV, while those on $\text{TiO}_x\text{N}_y/\text{SAM}$ reach 0.397% PLQY and 1.269 eV QFLS (Fig. 5g,h), confirming suppressed non-radiative recombination. Intensity-dependent QFLS measurements further reveal a reduced ideality factor ($n = 1.46$) for TiO_xN_y -based stacks compared with IZO-based counterparts ($n = 1.54$) (Fig. 5i). A lower ideality factor reflects the suppression of Shockley–Read–Hall (SRH) recombination, which typically originates from deep trap states at the SAM/perovskite interface⁴¹. These improvements can be attributed to two synergistic effects: (1) optimized energy alignment and surface potential homogeneity in the $\text{TiO}_x\text{N}_y/\text{SAM}$ facilitate more efficient electron-hole recombination through a reduced defect density; (2) stronger chemical interactions between TiO_xN_y and SAM enhance passivation and provide epitaxial templating during perovskite crystallization. Together, these mechanisms yield perovskite films with reduced defect density, improved optoelectronic quality and suppressed non-radiative recombination—factors that collectively enhance V_{oc} , FF and overall tandem efficiency.

Conclusions

We have developed TiO_xN_y as a multifunctional, indium-free RL for monolithic perovskite/Si tandem solar cells. TiO_xN_y addresses the limitations of TCO RLs and doped silicon tunnel junctions by combining high electrical conductivity, favourable band alignment and robust chemical bonding with SAMs. Our laboratory-size device and mini-module achieved high certified PCEs of 32.8% and 30.6%, respectively, with markedly enhanced operational stability compared to those devices based on IZO RLs. Mechanistic investigations reveal that TiO_xN_y provides abundant anchoring sites for SAMs, regulates perovskite crystallization and facilitates interfacial carrier recombination. These findings establish TiO_xN_y as a scalable and sustainable interconnection layer, offering a viable pathway towards the commercialization of perovskite/Si tandem photovoltaics.

Methods

Materials

All chemicals were used as received. Dimethyl sulfoxide (DMSO), dimethylformamide (DMF), isopropanol (IPA), ethanol and ethyl acetate were obtained from Sigma-Aldrich. FAI and MABr were obtained from GreatCell Solar Materials. C_{60} was purchased from Nano-C. CsI was purchased from Xi'an Yuri Solar. Piperazinium iodide (PiPI), ([4-(3,6-dimethoxy-9H-carbazol-9-yl)ethyl]phosphonic acid) (MeO-4PACz) and (4-(7H-dibenzo[c,g]carbazol-7-yl)butyl)phosphonic acid (4PACB) were purchased from Liwei technology. PbI_2 and PbBr_2 were purchased from TCI Chemicals. Tetrakis(dimethylamino)tin(IV) (TDMASn) was purchased from Ai Mou Yuan Scientific Equipment (Nanjing). Titanium target (99.999%), IZO target (In_2O_3 with 10 wt% ZnO), Ag and MgF_2 were purchased from Kunlang Advanced Material Technology Co. Ltd.

Silicon heterojunction bottom cells fabrication

N-type Cz silicon wafers (1 Ω cm) were etched in NaOH to remove surface damage, textured in diluted NaOH solution and subsequently subjected to RCA cleaning. Both wafer surfaces were passivated with an intrinsic hydrogenated amorphous silicon (a-Si(i), 5 nm) layer deposited by PECVD at 200 °C. An n-type hydrogenated nanocrystalline silicon oxide (nc- SiO_2 :H(n), 24 nm) layer was then deposited on the front side, whereas a p-type hydrogenated nanocrystalline silicon layer (nc-Si:H(p), 30 nm) was deposited on the rear side. An ITO transparent electrode (20 nm) was sputtered onto the rear surface, followed by screen printing of low-temperature silver paste and curing at 200 °C in air. An MgF_x layer (150 nm) and Ag electrode (500 nm) were subsequently deposited by thermal evaporation. After TiO_xN_y or IZO recombination layer deposition, the 6-inch wafers were then laser-cut into 25 mm $\times$ 25 mm cells. All thin-film deposition equipment was developed and manufactured by Suzhou Maxwell Technologies. The SHJ bottom cells were fabricated according to our previous report with minor modifications⁴².

TiO_xN_y and IZO recombination layers fabrication

The SHJ bottom cells were loaded in a magnetron sputtering chamber evacuated to 1×10^{-4} Pa (Hangzhou Zhongneng Photoelectricity Technology). A gas mixture of argon and nitrogen ($\text{Ar}/\text{N}_2 = 99:1$) was introduced into the chamber at a total gas flow rate of 15 sccm (standard cubic centimeters per minute). The titanium metal target was pre-sputtered at 200 W for 20 min, followed by TiO_xN_y deposition at the same power. The oxygen element in the film is primarily attributable to oxygen adsorbed during sample loading when the chamber is exposed to ambient air. This adsorbed oxygen subsequently desorbs during the deposition process and involved in various chemical reactions. IZO RL was sputtered in pure argon atmosphere (20 sccm) under a fixed power of 30 W and pressure of 5×10^{-4} Pa. The thickness of TiO_xN_y and IZO was tuned by varying sputtering time. An annealing process was performed in air at 200 °C for 30 min before perovskite top cell fabrication.

Tandem solar cells fabrication

For laboratory-scale tandem fabrication, a mixed SAM solution of 4PACB: MeO-4PACz (2:1 molar ratio, 1 mg ml^{-1}) in ethanol was spin coated onto the SHJ bottom cell at 4,000 rpm for 30 s and annealed at 100 °C for 10 min. A 1.7 M perovskite precursor solution of $\text{Cs}_{0.05}\text{FA}_{0.8}\text{MA}_{0.15}\text{Pb}(\text{I}_{0.75}\text{Br}_{0.25})_3$ was prepared in DMF/DMSO (4:1, v/v) and stirred before use. Perovskite films were deposited by two-step spin coating at 2,700 rpm for 55 s and 4,700 rpm for 10 s, with 400 μl ethyl acetate dripped 10 s before the end of spinning, followed by annealing at 100 °C for 20 min. PiPI (0.25 mg ml^{-1} in IPA) was then spin coated at 5,000 rpm for 30 s, annealed at 100 °C for 10 min and rinsed with IPA at the same speed. A 10-nm C_{60} layer was deposited by thermal evaporation (Hangzhou Zhongneng Photoelectricity Technology), followed by ALD SnO_2 buffer layer deposition (12 nm) at 90 °C using TDMASn and H_2O precursors (E200SP, SUPRALD). A 45-nm IZO electrode was sputtered, followed by thermal evaporation of a 600-nm Ag electrode. Finally, a 100-nm MgF_x anti-reflection layer was deposited at 0.1 nm s^{-1} . IZO reference devices were optimized under the same processing conditions and design constraints as the TiO_xN_y -based tandems. For industrial-size tandem fabrication, the process largely followed that used for laboratory-scale devices. A mixed 4PACB: MeO-4PACz (2:1 molar ratio, 0.2 mg ml^{-1}) in ethanol was slot-die coated onto SHJ bottom cells (210 mm $\times$ 105 mm) at 20 mm s^{-1} and 20 $\mu\text{l s}^{-1}$. The perovskite precursor solution was prepared as described for the laboratory-scale devices. Perovskite films were slot-die coated at 15 mm s^{-1} and 13 $\mu\text{l s}^{-1}$, followed by vacuum quenching at 100 mTorr for 20 s. PiPI (0.4 mg ml^{-1} in IPA) was subsequently slot-die coated at 15 mm s^{-1} and 15 $\mu\text{l s}^{-1}$. All annealing steps were identical to those used for laboratory-scale devices. C_{60} (10 nm) was then deposited by thermal evaporation, followed by ALD SnO_2 buffer layer (12 nm) deposition at 90 °C using TDMASn and H_2O . An IZO transparent electrode (45 nm) was sputtered at the front. Silver paste was screen-printed and annealed at 110 °C for 20 min. For tandem mini-module fabrication, 30 round busbars (200 μm diameter, 7 mm spacing) were attached to both sides of the device using thermoplastic polyolefin (TPO, 0.6 mm) and anti-reflection glass (2 mm) and finished by lamination at 110 °C for 20 min and butyl rubber edge sealing.

Perovskite and TiO_xN_y characterization

Scanning electron microscopy (SEM) was performed using a field-emission SEM (JSM-IT800, JEOL). XPS/UPS spectra were measured using a Kratos Axis Ultra XPS System. UPS measurements were carried out using a monochromatized He I (21.2 eV) source with -10 eV sample bias. XPS measurements used a He-ion UV source with a step size of 0.1 eV. Transmittance spectra of TiO_xN_y films (deposited on double-side polished quartz) were measured using a UV-Vis-NIR spectrophotometer with an integrating sphere. GIWAXS was carried out on a SaxsFocus system equipped with a Cu X-ray source (8.05 keV) and a Pilatus3R 300 K detector (0.5° incidence). XRD patterns were collected with a Rigaku SmartLab diffractometer. Time-resolved and steady-state photoluminescence (TRPL/PL) spectra were acquired using an Omni-Fluo-510HQ system with 450 nm excitation (fluence ~ 30 mJ cm^{-2}) and a 590 nm long-pass filter. KPFM was carried out using a Bruker Dimension Icon AFM. C-AFM was carried out using an AtomEdge Pro via the same type of tip, forward bias voltage of 1.5 V, and reverse bias of 1.5 V. (Truth Instruments Company, Qingdao). HAADF-STEM images and EDX mappings were obtained on an FEI Tecnai F30 TEM at 300 kV. PL mappings were performed using a Visiointec Luminescence Analyser under uniform illumination from two 530 nm lasers. Perovskite emission was imaged using a Princeton Instruments silicon CCD camera through a 730 ± 30 nm bandpass filter (100 ms exposure). Photoluminescence quantum yield (PLQY) was measured using a 532 nm laser at 1-sun equivalent intensity. The excitation light was delivered into the integrating sphere through an optical fibre to illuminate the sample, while the resulting emission was collected by a second fibre and analysed

with a spectrometer. Intensity dependent QFLS analysis and final fitting: plot the extracted QFLS (ΔE_F) versus the logarithm of the light intensity ($\log_{10} I$, where I is the excitation intensity normalized to 1-sun equivalent). Perform a linear fit using the equation: $\Delta E_F \approx nkT \ln(I/I_0) + c$, where k is Boltzmann's constant, T is absolute temperature, I_0 is a reference intensity, c is a constant.

Tandem solar cell characterization

J - V characteristics were recorded inside a nitrogen-filled glovebox using a Keithley 2400 source meter under illumination from a Wave-labs Sinus 300 LED solar simulator (AM1.5 G). The light intensity was calibrated with a Fraunhofer ISE-certified reference cell. Voltage sweeps were performed from -0.02 V to 2 V with a step size of 0.02 V and an integration/settling time of 20 ms. The active area (1 cm^2) was defined by a shadow mask. R_{sh} was characterized using perovskite single-junction test cells (1 cm^2) fabricated on highly conductive textured silicon wafers ($1 \Omega \cdot \text{cm}$) as Ohmic substrates, following identical RL and perovskite processing conditions as the full tandem devices. A metal contact was applied to the rear surface to enforce purely vertical charge transport through the wafer, replicating the vertical carrier recombination pathway in a monolithic tandem cell. R_{sh} was extracted from dark J - V measurements as the ratio of terminal voltage to current within the measured voltage range. EQE spectra were measured using a Bentham EQE system. For the top perovskite subcell, measurements were performed with near-infrared bias light (~ 845 nm LED, intensity equivalent to ~ 1 sun for the Si bottom cell) with 0.5 V bias voltages. Conversely, for the Si bottom subcell, green bias light (~ 525 nm LED, ~ 1 sun equivalent for the perovskite) was applied without external voltage bias, isolating the Si response. Probe light intensity was maintained at low levels (~ 0.01 sun equivalent broadband). The active area (1 cm^2) was defined by a shadow mask. Suns- V_{oc} measurements were performed to extract pseudo J - V parameters. The selective characterization of subcells was enabled through a wavelength-specific bias illumination protocol. The silicon bottom cell was saturated using 900 nm infrared bias light to permit exclusive characterization of the perovskite top cell, whereas the perovskite top cell was saturated using 450 nm blue bias light to allow exclusive characterization of the silicon bottom cell. Under identical current-matched spectral conditions, independent J_{sc} and suns- V_{oc} curves were thereby obtained for each subcell. These individually determined J_{sc} values were then employed to correctly scale the respective suns- V_{oc} data, generating pseudo J - V curves for each subcell. Finally, the ideal tandem performance curve was reconstructed through voltage summation of these subcell curves. For the long-term stability test, the laboratory-size tandems were sandwiched between two 3 -mm-thick cover glass/encapsulant (polyolefin elastomer, POE) with black butyl rubber sealant at the edges. The device was subjected to vacuum lamination in an industrial laminator at a temperature of 120°C for a duration of 20 mins. Copper ribbons were utilized to establish contact between the front and rear electrodes of the tandems, with these strips being extended to the exterior of the cover glass. To evaluate the impact of RLs on device stability, both laboratory-size and industrial-size tandems without Pipl passivation were deployed for all the stability test, and all other conditions were maintained except for the use of the TiO_xN_y and IZO RLs.

Stability testing for laboratory-size solar cells

Maximum power point (MPP) tracking (ISOS-L-II): the devices were operated under simulated sunlight at 1-sun intensity in a nitrogen atmosphere with continuous MPP tracking. The average cell temperature was maintained at approximately 25°C by active cooling throughout the test.

Damp-heat test (ISOS-D-3): the devices were aged in an environmental chamber at 85°C and 85% relative humidity. At predefined time intervals, standard J - V measurements were performed to evaluate performance degradation.

Stability testing for industrial-size solar cells

MPP tracking (ISOS-L-2I): the devices were exposed to simulated sunlight with continuous MPP tracking in an inert atmosphere. The average cell temperature was controlled at approximately 85°C to assess the combined effects of thermal and illumination stress.

Damp-heat test (ISOS-D-3): the test conditions were identical to those applied to the small-area devices, ageing at 85°C and 85% relative humidity, with periodic J - V measurements to monitor performance evolution.

Computational methods

The SPICE simulation represents a tandem solar cell model consisting of subcells. The models used for the two perovskite cells are pvs1 and pvs2. The first perovskite cell (pvs1) is modelled as a diode with a saturation current $I_s = 2 \times 10^{-23}$ A and an ideality factor $N = 1$. The pvs2 has $I_s = 2 \times 10^{-13}$ A and $N = 2$. The first silicon cell (c-Si1) with $I_s = 9.8 \times 10^{-15}$ A and $N = 1$. The c-Si2 has $I_s = 1 \times 10^{-9}$ A and $N = 2$. DC sweep of the voltage source (V1) from 0 V to 2 V in steps of 0.001 V. The $R_s = 0.5$ ohms. R_{sh1} for the silicon cell and R_{sh2} for the perovskite cell.

The band diagrams of the tandem were calculated using SCAPS-1D. For each layer, user-defined parameters include the thickness, bandgap (E_g), electron affinity (E_A), dielectric permittivity (ϵ), conduction band and valence band effective density of states (N_c and N_v), electron and hole thermal velocity (v_{thn} and v_{thp}), electron and hole mobility (μ_n and μ_p), donor and acceptor density (N_D and N_A), bulk defect type and distribution and the interface defect type and distribution. The complete set of parameters used to generate the band diagrams is summarized in Supplementary Tables 3–5. SCAPS-1D then numerically solves the coupled system of Poisson's equation (1), the continuity equations (2) and (3) and the drift-diffusion equations (4) and (5) at steady state⁴³:

$$\frac{\partial}{\partial x} \left(\epsilon_0 \epsilon \frac{\partial \Psi}{\partial x} \right) = -q \left(p - n + N_D^+ - N_A^- + \frac{\rho^{\text{def}}}{q} \right) \quad (1)$$

$$-\frac{\partial J_n}{\partial x} - U_n + G = \frac{\partial n}{\partial t} \quad (2)$$

$$-\frac{\partial J_p}{\partial x} - U_p + G = \frac{\partial p}{\partial t} \quad (3)$$

$$J_n = -\frac{\mu_n n}{q} \frac{\partial E_{Fn}}{\partial x} \quad (4)$$

$$J_p = -\frac{\mu_p p}{q} \frac{\partial E_{Fp}}{\partial x} \quad (5)$$

where Ψ is the electrostatic potential, N_D^+ and N_A^- are the ionized donor and acceptor density, ρ^{def}/q is the user-defined defect/charge density, J_n and J_p are current densities, U_n and U_p are recombination rates, G is the generation rate, p and n are the hole density and electron density, and E_{Fn} and E_{Fp} are the quasi-Fermi levels.

Geometry optimizations for both bulk and surface adsorption were performed with the Vienna Ab Initio Simulation Package (VASP, version 6.4.2)⁴⁴ using projector-augmented wave pseudopotentials and the PBE exchange-correlation functional within the generalized gradient approximation. Van der Waals dispersion interactions were included via the DFT-D3 scheme⁴⁵. Gamma-centred grids were employed. A $3 \times 3 \times 3$ and a $9 \times 3 \times 3$ k -point mesh was used for the bulk structures of In_2O_3 (space group: Ia $\bar{3}$) and $\text{Ti}_3\text{O}_5\text{N}_2$ (space group: cmcm), respectively²². Zn (5%) were doped under furthest distance considerations to form IZO. Slab dimensions of the adsorption surface were sized at $9.82 \times 9.91 \text{ \AA}^2$ (6 atomic layers) and $10.21 \times 10.21 \text{ \AA}^2$ (five atomic layers) for $\text{Ti}_3\text{O}_5\text{N}_2$ and IZO, respectively. Both structures were exposed with the (100) facet and a $15 \text{ \AA}$ of vacuum layer to prevent periodicity

interference. A $3 \times 3 \times 1$ k-point mesh grid was employed for the adsorption interfaces of SAM/Ti₃O₃N₂ and SAM/IZO (SAM: 4PADCB/MeO-4PACz), respectively. The plane-wave basis energy cut-offs were set at 450 eV. The convergence criteria of energy and force were set to 1×10^{-5} eV and -0.015 eV Å⁻¹, respectively. The dipole corrections were also included in the work function calculations for the slab models. The binding energy of interfacial models were calculated using the following equation of $E_b = E_{\text{tot}} - E_{\text{slab}} - E_{\text{mol}}$, where E_{tot} is the total energy of the relaxed configurations for SAM/IZO and SAM/Ti₃O₃N₂ cases; E_{slab} is the energy of IZO (100) and Ti₃O₃N₂ (100) slabs; and E_{mol} is the energy of the SAM molecule in respective cases. The LOBSTER package was used for analysing the COHP and ICOHP bond interactions, with Fermi level levelled to origin for ICOHP analysis⁴⁶.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data are available in the main text or the supplementary materials. Source data are provided with this paper.

Code availability

Vienna Ab initio Simulation Package (VASP) for the DFT calculations is available at <https://www.vasp.at/>.

References

- Han, J. et al. Perovskite solar cells. *Nat. Rev. Methods Primers* **5**, 3 (2025).
- Zhang, X. et al. Advances in inverted perovskite solar cells. *Nat. Photonics* **18**, 1243–1253 (2024).
- Mei, J. & Yan, F. Recent advances in wide-bandgap perovskite solar cells. *Adv. Mater.* **37**, 2418622 (2025).
- Green, M. A. et al. Solar cell efficiency tables: version 68. *Joule* **10**, 102494 (2026).
- Photovoltaic research. Best research-cell efficiency chart. *NLR* <https://www.nlr.gov/pv/cell-efficiency> (2026).
- Shen, H. et al. Monolithic perovskite/Si tandem solar cells: pathways to over 30% efficiency. *Adv. Energy Mater.* **10**, 1902840 (2020).
- Fu, F. et al. Monolithic perovskite-silicon tandem solar cells: from the lab to fab? *Adv. Mater.* **34**, 2106540 (2022).
- Chin, X. Y. et al. Interface passivation for 31.25%-efficient perovskite/silicon tandem solar cells. *Science* **381**, 59–63 (2023).
- Al-Ashouri, A. et al. Monolithic perovskite/silicon tandem solar cell with >29% efficiency by enhanced hole extraction. *Science* **370**, 1300–1309 (2020).
- Leijtens, T., Bush, K. A., Prasanna, R. & McGehee, M. D. Opportunities and challenges for tandem solar cells using metal halide perovskite semiconductors. *Nat. Energy* **3**, 828–838 (2018).
- Dong, C. et al. Principle and progress of interconnection layers in monolithic perovskite-based tandem photovoltaics. *Adv. Energy Mater.* **15**, 2404628 (2025).
- Ko, Y., Park, H., Lee, C., Kang, Y. & Jun, Y. Recent progress in interconnection layer for hybrid photovoltaic tandems. *Adv. Mater.* **32**, 2002196 (2020).
- Shao, Y. et al. Innovations in interconnecting layers for perovskite-based tandem solar cells. *ACS Energy Lett.* **9**, 4892–4921 (2024).
- De Bastiani, M. et al. Recombination junctions for efficient monolithic perovskite-based tandem solar cells: physical principles, properties, processing and prospects. *Mater. Horiz.* **7**, 2791–2809 (2020).
- Aydin, E. et al. Enhanced optoelectronic coupling for perovskite/silicon tandem solar cells. *Nature* **623**, 732–738 (2023).
- Chi, W. et al. Perovskite/silicon tandem solar cells: choice of bottom devices and recombination layers. *ACS Energy Lett.* **8**, 1535–1550 (2023).
- Zheng, J. et al. Polycrystalline silicon tunnelling recombination layers for high-efficiency perovskite/tunnel oxide passivating contact tandem solar cells. *Nat. Energy* **8**, 1250–1261 (2023).
- Aydin, E. et al. Pathways toward commercial perovskite/silicon tandem photovoltaics. *Science* **383**, eadh3849 (2024).
- Köhnen, E. et al. Highly efficient monolithic perovskite silicon tandem solar cells: analyzing the influence of current mismatch on device performance. *Sustain. Energy Fuels* **3**, 1995–2005 (2019).
- Yang, X. et al. A highly conductive titanium oxynitride electron-selective contact for efficient photovoltaic devices. *Adv. Mater.* **32**, 2002608 (2020).
- Peng, J. et al. Centimetre-scale perovskite solar cells with fill factors of more than 86 percent. *Nature* **601**, 573–578 (2022).
- Rees, K. et al. Combining single source chemical vapor deposition precursors to explore the phase space of titanium oxynitride thin films. *Dalton Trans.* **47**, 10536–10543 (2018).
- Sowińska, M. et al. Plasma-enhanced atomic layer deposition of titanium oxynitrides films: a comparative spectroscopic and electrical study. *J. Vac. Sci. Technol. A* **34**, 01A127 (2015).
- Sowińska, M. et al. Analysis of nitrogen species in titanium oxynitride ALD films. *Appl. Surf. Sci.* **381**, 42–47 (2016).
- Yang, X. et al. Dual-function electron-conductive, hole-blocking titanium nitride contacts for efficient silicon solar cells. *Joule* **3**, 1314–1327 (2019).
- Wang, Y., Wang, Y., Gao, F. & Yang, D. Efficient monolithic perovskite/silicon tandem photovoltaics. *Energy Environ. Mater.* **7**, e12639 (2024).
- Wright, M. et al. Design considerations for the bottom cell in perovskite/silicon tandems: a terawatt scalability perspective. *Energy Environ. Sci.* **16**, 4164–4190 (2023).
- Ma, Y., Gong, J., Zeng, P. & Liu, M. Recent progress in interfacial dipole engineering for perovskite solar cells. *Nano Micro Lett.* **15**, 173 (2023).
- Angus, F. J. et al. Understanding the impact of SAM fermi levels on high efficiency p-i-n perovskite solar cells. *J. Phys. Chem. Lett.* **15**, 10686–10695 (2024).
- He, R. et al. Improving interface quality for 1-cm² all-perovskite tandem solar cells. *Nature* **618**, 80–86 (2023).
- Ren, Z. et al. Poly(carbazole phosphonic acid) as a versatile hole-transporting material for p-i-n perovskite solar cells and module. *Joule* **7**, 2894–2904 (2023).
- Paniagua, S. A. et al. Phosphonic acids for interfacial engineering of transparent conductive oxides. *Chem. Rev.* **116**, 7117–7158 (2016).
- Paramonov, P. B. et al. Theoretical characterization of the indium tin oxide surface and of its binding sites for adsorption of phosphonic acid monolayers. *Chem. Mater.* **20**, 5131–5133 (2008).
- Qu, G. et al. Conjugated linker-boosted self-assembled monolayer molecule for inverted perovskite solar cells. *Joule* **8**, 2123–2134 (2024).
- Jiang, W. et al. π -expanded carbazoles as hole-selective self-assembled monolayers for high-performance perovskite solar cells. *Angew. Chem. Int. Ed.* **61**, e202213560 (2022).
- Tang, H. et al. Reinforcing self-assembly of hole transport molecules for stable inverted perovskite solar cells. *Science* **383**, 1236–1240 (2024).
- Li, Y. et al. Ultra-high open-circuit voltage of perovskite solar cells induced by nucleation thermodynamics on rough substrates. *Sci. Rep.* **7**, 46141 (2017).
- Yang, Y. et al. Synchrotron radiation-based in situ GIWAXS for metal halide perovskite solution spin-coating fabrication. *Adv. Sci.* **11**, 2403778 (2024).

39. Krogmeier et al. Quantitative analysis of the transient photoluminescence of $\text{CH}_3\text{NH}_3\text{PbI}_3/\text{PC}_{61}\text{BM}$ heterojunctions by numerical simulations. *Sustain. Energy Fuels* **2**, 1027–1034 (2018).
40. Luo, C. et al. Engineering bonding sites enables uniform and robust self-assembled monolayer for stable perovskite solar cells. *Nat. Mater.* **24**, 1265–1272 (2025).
41. Sarritzu, V. et al. Optical determination of Shockley-Read-Hall and interface recombination currents in hybrid perovskites. *Sci. Rep.* **7**, 44629 (2017).
42. Wang, S. et al. Flexible perovskite/silicon tandem solar cells with 33.6% efficiency. *Nature* **649**, 59–64 (2026).
43. Burgelman, M. et al. *SCAPS Manual* (University of Ghent, Department of Electronics and Information Systems, 2025).
44. Kresse, G. & Furthmüller, J. Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set. *Phys. Rev. B* **54**, 11169–11186 (1996).
45. Grimme, S. Density functional theory with London dispersion corrections. *WIREs Comput. Mol. Sci.* **1**, 211–228 (2011).
46. Nelson, R. et al. LOBSTER: local orbital projections, atomic charges, and chemical-bonding analysis from projector-augmented-wave-based density-functional theory. *J. Comput. Chem.* **41**, 1931–1940 (2020).

Author contributions

X.Y. conceived the idea, designed the overall experiments and led the project. X.Y. and F.C. wrote the paper. F.C., Y.L. and S.W. fabricated the tandem devices and performed the characterization and analysis. K.L.W. and J. Yin performed the DFT calculations. C.Y., S.Y., X.C. and J. Zhou fabricated the SHJ bottom cells. X.N. and R.S.B. performed the SCAPS-1D simulations. B.G., W.L., W.S., K.G., L.Y. and B.Y. contributed to characterizations and data interpretation. J. Yu, Y.Z., Y.W. and J. Zhu fabricated the tandem mini-module. W.A., S.D. and L.S. helped the characterization and thin films deposition. F.W. and H.H. performed the GIWAXS measurements. X.Z. and X.Y. supervised this project. All authors contributed to the discussion of the results and revision of the paper.

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Competing interests

J. Zhou, C.Y., S.Y. and X.C. are employees of Suzhou Maxwell Technologies Co. Ltd. J. Yu, Y.Z., Y.W. and J. Zhu are employees of Wuxi EliTe Solar Co. Ltd. W.A., S.D. and L.S. are employees of Hangzhou Zhongneng Photoelectricity Technology Co. Ltd. The other authors declare no competing interests.

Additional information

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Solar Cells Reporting Summary

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► Experimental design

Please check the following details are reported in the manuscript, and provide a brief description or explanation where applicable.

1. Dimensions

Area of the tested solar cells

☒ Yes
☐ No

The active area of the small-size device is 0.9949 cm², and that of the mini-module device is 207.87 cm².

Explain why this information is not reported/not relevant.

Method used to determine the device area

☒ Yes
☐ No

Device characterization in Method

Explain why this information is not reported/not relevant.

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction

☒ Yes
☐ No

In Fig. 1c and Fig. 2a .

Voltage scan conditions

☒ Yes
☐ No

Current-voltage (J-V) curves were obtained with a step size of 20 mV and a delay time of 20 ms.

Explain why this information is not reported/not relevant.

Test environment

☒ Yes
☐ No

At 25 °C air environment.

Explain why this information is not reported/not relevant.

Protocol for preconditioning of the device before its characterization

☒ Yes
☐ No

No any preconditioning was used in this work.

Explain why this information is not reported/not relevant.

Stability of the J-V characteristic

☒ Yes
☐ No

In Fig. 1d and Fig. 2b .

Explain why this information is not reported/not relevant.

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization

☒ Yes
☐ No

Our solar cells show minimal hysteresis.

Explain why this information is not reported/not relevant.

Related experimental data

☒ Yes
☐ No

In Fig. 1c and Fig. 2a .

Explain why this information is not reported/not relevant.

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)

☒ Yes
☐ No

In Fig. 1e, Fig. 2c and Supplementary Fig.29.

Explain why this information is not reported/not relevant.

A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator

☒ Yes
☐ No

From the external quantum efficiency (EQE) spectra , the integrated Jsc are equivalent to the values acquired from the J-V curves.

Explain why this information is not reported/not relevant.

For tandem solar cells, the bias illumination and bias voltage used for each subcell	☒ Yes ☐ No	Device characterization in Method. <i>Explain why this information is not reported/not relevant.</i>
5. Calibration		
Light source and reference cell or sensor used for the characterization	☒ Yes ☐ No	The reference cell was certified by the Fraunhofer ISE. <i>Explain why this information is not reported/not relevant.</i>
Confirmation that the reference cell was calibrated and certified	☒ Yes ☐ No	The reference cell was certified by the Fraunhofer ISE. <i>Explain why this information is not reported/not relevant.</i>
Calculation of spectral mismatch between the reference cell and the devices under test	☒ Yes ☐ No	The mismatch factor for both cells is around 1.00+0.01. The spectra irradiance in our lab was measured regularly using an external spectrometer. <i>Explain why this information is not reported/not relevant.</i>
6. Mask/aperture		
Size of the mask/aperture used during testing	☒ Yes ☐ No	The aperture area is determined by a certification laboratory. <i>Explain why this information is not reported/not relevant.</i>
Variation of the measured short-circuit current density with the mask/aperture area	☒ Yes ☐ No	Japan Electrical Safety & Environment Technology Laboratories calibrated the small-size mask area, in Supplementary Fig.3. National PV Industry Measurement and Testing Center (NPVM) calibrated the mini-module area, in the Supplementary Fig.14. <i>Explain why this information is not reported/not relevant.</i>
7. Performance certification		
Identity of the independent certification laboratory that confirmed the photovoltaic performance	☒ Yes ☐ No	Japan Electrical Safety & Environment Technology Laboratories confirmed the laboratory-size device performance and NPVM confirmed the mini-module device performance. <i>Explain why this information is not reported/not relevant.</i>
A copy of any certificate(s)	☒ Yes ☐ No	In Supplementary Fig.3 and Supplementary Fig.14. <i>Explain why this information is not reported/not relevant.</i>
8. Statistics		
Number of solar cells tested	☒ Yes ☐ No	In the main body and supplemental information of this work, 20+20+20+20= 80 devices <i>Explain why this information is not reported/not relevant.</i>
Statistical analysis of the device performance	☒ Yes ☐ No	In Fig. 3e and Supplementary Fig.25, Supplementary Fig.4,Supplementary Fig.16 and Supplementary Fig.32. <i>Explain why this information is not reported/not relevant.</i>
9. Long-term stability analysis		
Type of analysis, bias conditions and environmental conditions	☒ Yes ☐ No	Device characterization was performed following the procedures detailed in the Method, and the corresponding results are depicted in Fig. 1f, Fig. 1g, Fig. 2d, and Fig. 2e. <i>Explain why this information is not reported/not relevant.</i>